

Knowledge Check 1

Name: _____	Section: _____
Date: _____	Score: _____ / 20

Instructions: Complete this written quiz individually during class. Circle the best answer for each question. You may use printed or handwritten course materials. No devices or technological assistance are permitted.

- Analytics is about using data to _____ managerial decision-making.
 - drive
 - inform
 - frame
- In the West, 120 of 600 customers belong to a campaign's target audience. In the South, 80 of 200 customers belong to that audience. Which statement is correct?
 - The West has the larger target audience; the South has the higher target-audience percentage.
 - The South has the larger target audience; the West has the higher target-audience percentage.
 - The West has both the larger target audience and the higher target-audience percentage.
- `customer\_data` contains 400 customers across four regions, with no missing income values. What does this code return?

```
customer_data |>
  group_by(region) |>
  summarize(avg_income = mean(income))
```

 - 400 rows, with each customer's regional average income added.
 - Four rows, with one average income for each region.
 - One row, with the average income across all customers.
- What symbol needs to be placed at the beginning of a comment in a chunk of R code?
 - |>
 - \$
 - #

5. A customer table contains 100 customers, one row per customer. A December activity table contains one row for each of the 60 customers who purchased that month. You need a report containing all 100 customers, with December activity attached where available. Which approach produces that report?

- A. Left join the activity table onto the customer table using `customer\_id`.
- B. Inner join the two tables using `customer\_id`.
- C. Left join the customer table onto the activity table using `customer\_id`.

6. Two regions have the same average customer income. Management wants to know whether incomes are similarly distributed within the regions. Which visualization best addresses that question?

- A. A bar chart showing average income for each region.
- B. Side-by-side boxplots of individual customer incomes by region.
- C. A bar chart showing the number of customers in each region.

7. In a data frame, rows are called _____ columns are called _____.

- A. observations, variables
- B. variables, observations
- C. vectors, respondents

8. A customer table includes a binary variable, `college\_degree`, coded 1 if a customer has earned a college degree and 0 otherwise. What information does `college\_degree` contain?

- A. The name of the highest degree the customer has earned.
- B. The name of the university the customer attended.
- C. An indicator of whether the customer has a college degree.

9. Suppose `customer\_data` contains one row per customer and a `state` column. Which pipeline correctly counts the customers whose state is New Mexico?

- A. `customer_data |> filter(state == "New Mexico") |> summarise(n = n())`
- B. `customer_data |> select(state == "New Mexico") |> summarise(n = n())`
- C. `customer_data |> filter(state == "New Mexico") |> mutate(n = n())`

10. What is the correct way to create a new data frame composed of customer_id, region, and income using customer_data?

- A. `select(customer_data, customer_id, region, income)`
- B. `filter(customer_data, customer_id, region, income)`
- C. `slice(customer_data, customer_id:income)`

11. An inner join keeps all rows in the “left” data frame plus all columns from both data frames.
- A. True
 - B. False
12. Which pipeline returns one row per region, with the number of customers in each region ordered from smallest to largest?
- A. `customer_data |> count(region) |> arrange(n)`
 - B. `customer_data |> count(region) |> arrange(desc(n))`
 - C. `customer_data |> arrange(region) |> count(region)`
13. What are the three required layers (in order) in the grammar of graphics?
- A. aesthetics, geom, facets
 - B. geom, aesthetics, data
 - C. data, aesthetics, geom
14. You want to tokenize your text (currently in a column named “text”) using `unnest_tokens()`. Which of the following code snippets will work?
- A. `data |> unnest_tokens(word, text)`
 - B. `data |> unnest_tokens(text, word)`
 - C. `unnest_tokens(data, word, texts)`
15. You are exploring the relationship between two continuous variables where overplotting is an issue. Which of the following geoms should you use?
- A. `geom_col()`
 - B. `geom_point()`
 - C. `geom_jitter()`
16. Customer ZIP codes were imported as numbers. You want to compare income distributions across ZIP-code groups. How should the ZIP-code variable be treated?
- A. As categories, using a character or factor column, because ZIP codes identify groups.
 - B. As a continuous measurement, because the ZIP codes contain digits.
 - C. As a count, using an integer column, because ZIP codes contain no decimal places.
17. Following the tidy data philosophy generally results in longer data sets.
- A. True
 - B. False

18. Suppose `store_transactions` contains one row per transaction and includes `customer_id` and `transaction_date`. Which pipeline returns the customer with the most transactions during 2007?

- A. `store_transactions |> filter(year(transaction_date) == 2007) |> count(customer_id, sort = TRUE) |> slice_head(n = 1)`
- B. `store_transactions |> filter(year(transaction_date) == 2007) |> count(customer_id) |> slice_head(n = 1)`
- C. `store_transactions |> count(year(transaction_date), sort = TRUE) |> slice_head(n = 1)`

19. You have dataframe called `location`. One of the columns in that data frame is called `city_state`, which includes city and state separated by a comma. Which code snippet will correctly split the city and state into two columns?

- A. `location |> separate(city_state, c(city, state), sep = ",")`
- B. `separate(location, city_state, c("city", "state"), sep = ",")`
- C. `pivot_wider(location, city_state, c("city", "state"), sep = ",")`

20. If you have dates in a character column (e.g., "22-Oct-2012"), which `{lubridate}` function will allow you to coerce the column into type date?

- A. `mdy()`
- B. `dmy()`
- C. `ymd()`